

does offer two unique things – the guest OS runs within the console, instead of a separate instance, and the support for audio devices and video cards is terrific. For example, in this screenshot, Ubuntu is being run within VMware Console, rather than a separate window, as would have been the case with VirtualBox. This gives the product an organised appeal.

General use

Before we proceed any further to each product's performance levels, it is worth noting that Parallels Workstation is a software meant essentially for the server environment, whereas VMware Player is meant for non-enterprise use. The other two, Qemu and VirtualBox, however, can double up for both scenarios.

Qemu, strictly speaking, does not project itself as virtualisation software. It is more of an emulation mechanism for the x86 architecture. Well, there are certain scenarios where it does perfectly well, and beats all others. For instance, you will not need to tweak sound profiles to make them work on the guest OS. The downside, however, is that Qemu's guest machines, even when idle, tend to consume a good deal of memory.

VirtualBox, on the other hand, strikes a great balance between memory consumption and hardware detection. You will need to make a few extra tweaks at times to get the audio working, but overall, the software is wonderful. Even when running memory hogging operating systems such as Windows Vista, you will rarely see a performance lag (assuming that you have a decent RAM in place).

Parallels Workstation, since it is basically meant for workstation use, detects any Internet connection that you may have, by default (if you do not wish to share the host connection with the guest machine, you will have to disable it manually). The detection of sound devices is ordinary, and running videos is a pain. The program itself eats up a good deal of memory.

VMware Player is, by far, one of the most impressive virtualisation tools when it comes to performance. In my experience, there has rarely ever been a performance lag. It automatically detects many hardware devices, and memory management is great!



Note: Certain screenshots are courtesy of Wikimedia Commons.

Now, going beyond features, here is a quick view on the performance levels of each option: I attempted to install Mandriva LXDE under each virtualisation tool (the host OS being Sabayon 8, the RAM available to the guest OS was 256 MB, and storage space was 6 GB). In Qemu, the entire installation process took roughly 20 minutes; VirtualBox was around 15 minutes; VMware Player, too, stayed closer to 20; whereas Parallels Workstation took a little over half an hour. Seems comparable, doesn't it?

Appraisal

So, we began this article comparing four separate virtualisation tools. While you may have your personal preferences, in my opinion, Parallels Workstation, the costliest of the lot, fared in the most unimpressive manner (comparatively). On the other hand, VMware Player is a decent choice for virtualisation. And once you get used to using Qemu, it will serve you well. However, in terms of a combination of speed, ease of use and performance, VirtualBox ranked ahead of the lot. For all general purposes, VirtualBox will easily suffice, and just in case you need slightly advanced functionality, you should consider opting for VMware Player. 

Links

- Qemu: www.qemu.org
- VirtualBox: www.virtualbox.org
- VMware Player: <http://www.vmware.com/products/player/>
- Parallels Workstation: <http://www.parallels.com/products/workstation/>

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